

CHAPTER 8

Quality Control

8.1. Introduction¹

Quality is judged by the customer. It is meeting and exceeding customer needs. It is delighting the customer. It is customer satisfaction. Quality sounds so simple, but only the best organizations worldwide are making these things happen on a day in, day out basis.

Quality and customer satisfaction have become the key drivers behind some of the most profound changes that have occurred in the world of work in all of recorded history. The focus today is on the customer, recognizing that there are many customers. For example, there are the external customers who buy and use a product or service. There are the internal customers who are the next persons in the process, perhaps working right next to us. We are trying to meet or exceed the needs of *all* of our customers today.

The emphasis on quality has not been an isolated event. It is intertwined with virtually all of industrial engineering. For example, JIT and MRP (Chapter 7) will simply not work unless quality of products, deliveries, record keeping, and so on,

¹Much of the content of this chapter comes from the materials of Exxon Chemical Company, Associates in Process Improvement, and from the Malcolm Baldrige National Quality Award Office of the National Institute of Standards and Technology, U.S. Department of Commerce.

from suppliers is top notch. Quality and JIT have affected facilities design, requiring that workstations be placed close together or even in cells, placing the internal supplier and internal customer elbow to elbow. Similar examples could go on and on.

8.2. A Bit of History

During the twentieth century, there has been a big flurry of ideas put forth under names such as quality control, statistical process control, designed experimentation, quality planning, quality costs, reliability, quality circles, zero defects, quality audit, quality function deployment, total quality management, benchmarking, and others. Each of these topics and many others have generated enthusiasm, research, and implementation. Each was successful in various places; similarly, each failed in other places.

Companies have evolved. After World War II, those manufacturers not severely affected by the war (like those in the United States) had little global competition. High short-term profitability was achieved by high production levels. Many manufacturing companies neglected quality in the push for production. Quality suffered, but manufacturers saw little reason to change since markets and profits continued.

We all know that the situation has changed. Global competition is intense. Industries and companies are faced with closings, sectors of high unemployment, and the loss of many high-paying jobs. Companies around the world that sacrifice quality improvement for increased production are in deep trouble. This is because a "world class" breed of companies now achieve both high quality and high productivity at lower cost. *No industry or company is immune to this new competition.*

Fortunately, we continue to learn what common principles and tools make some ideas work and some companies successful. Some of these common principles and tools are detailed in this chapter.

8.3. The Malcolm Baldrige National Quality Award²

The single most dramatic event in the past decade to foster quality in the United States (and even worldwide) has been the offering of the Malcolm Baldrige National Quality Award. The criteria for the award are extensive and they give a de facto definition of the words *total quality management* or (TQM). The specific topics addressed by the criteria fall into the seven categories shown in Figure 8.1. The seven major categories of the Baldrige criteria are often illustrated as shown in

²Kenneth E. Case has served on the panel of judges and also as a senior examiner in the Baldrige process.

1.0 LEADERSHIP

Senior Executive Leadership
Management for Quality
Public Responsibility

2.0 INFORMATION AND ANALYSIS

Scope and Management of Quality and Performance Data and Information
Competitive Comparisons and Benchmarks
Analysis and Uses of Company-Level Data

3.0 STRATEGIC QUALITY PLANNING

Strategic Quality and Company Performance Planning Process
Quality and Performance Plans

4.0 HUMAN RESOURCE DEVELOPMENT AND MANAGEMENT

Human Resource Management
Employee Involvement
Employee Education and Training
Employee Performance and Recognition
Employee Well-Being and Morale

5.0 MANAGEMENT OF PROCESS QUALITY

Design and Introduction of Quality Products and Services
Process Management-Product and Service Production and Delivery Processes
Process Management-Business Processes and Support Services
Supplier Quality
Quality Assessment

6.0 QUALITY AND OPERATIONAL RESULTS

Product and Service Quality Results
Company Operational Results
Business Process and Support Service Results
Supplier Quality Results

7.0 CUSTOMER FOCUS AND SATISFACTION

Customer Relationship Management
Commitment to Customers
Customer Satisfaction Determination
Customer Satisfaction Results
Customer Satisfaction Comparison
Future Requirements and Expectations of Customers

Fig. 8.1. Malcolm Baldrige National Quality Award Categories/Items.

Figure 8.2. Companies are using these criteria to measure themselves over time and against others, both in their own industry and in other industries. The primary motivation for this is to continuously improve, increase competitiveness, capture the market, survive, and thrive.

There are several “bedrock quality concepts” underlying the MBNQA criteria and the worldwide quality movement today. Interestingly, they are all also bedrock concepts underlying the practice of industrial engineering.

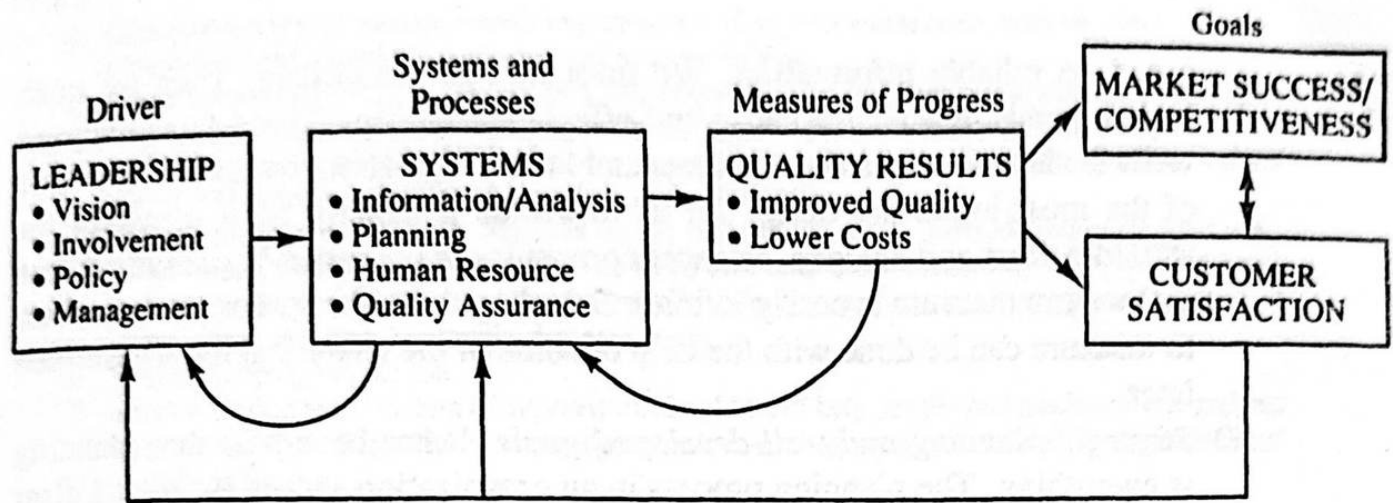


Fig. 8.2. National quality award categories dynamic relationships.

- (1) *Quality is customer driven.* Since quality is judged by the customer, all product and service attributes that contribute value to the customer and affect customer preference must be considered in a quality system. Customer-driven quality demands constant awareness of changing customer wants and needs and rapid response to meet these requirements.
- (2) *Leadership creates quality values.* A company's senior leaders must create clear quality values, specific goals, and well-defined systems and methods for achieving the goals. These systems and methods need to guide all activities of the company and to encourage participation by all employees. Senior leaders can make or break a company.
- (3) *Optimized systems and processes.* Quality excellence comes from well-defined and well-executed work systems and processes (this is exactly what industrial engineers are concerned with). Quality is achieved by focusing on systems and processes and doing the right things right the first time. It is not achieved by focusing on inspection and correction of products or services after they have been made or delivered.
- (4) *Response-time reduction.* Meeting changing customer requirements and expectations in competitive markets demands shorter cycles of concept-design-build-deliver. For example, an automaker that can deliver a new concept car in three years will far outdistance a car company that requires four or five years. Reducing response time often occurs when work processes are simplified and shortened, usually also resulting in improved quality.
- (5) *Continuous improvement.* Improvements to all operations and work activities may be of several types: (a) increasing value to the customer by improving product features, (b) reducing errors and defects, (c) improving responsiveness and cycle time performance, and (d) improving efficiency and effectiveness in the use of all resources.
- (6) *Actions based on facts, data, and analysis.* Real improvement requires that actions be taken to change and improve processes. These changes must be

based on reliable information. We must first get good data. Then we must look at trends, levels, and cause-and-effect relationships. Good data, analyzed well, leads to facts, understanding, and knowledge about our processes. One of the most important things for us to decide is exactly what measures we should collect and analyze. In most companies the number of possible things that we can measure is nearly infinite. Selecting those things that are important to measure can be done with the help of some of the analytical tools described later.

- (7) *Strategic planning and well-developed goals.* It has been said that planning is everything. The planning process in an organization should be crystal clear to everyone. Employees at all levels should be able to imput to the planning effort. The major goals should lay out a well-defined direction for the company. The plan should then cascade through the company, with specific activities for individuals and teams at all levels.
- (8) *Employees trained, developed, and involved.* Real progress toward world class requires that all employees be committed, well trained, and encouraged to participate in continuous improvement activities. The reward and recognition systems must be aligned with the goals of the company—world class will be impossible to achieve if the company rewards employees for “business as in the past.” Safety, health, well-being, and morale of employees are an integral part of getting them involved in continuous improvement.
- (9) *Supplier involvement.* Incoming products and services must be of high quality if we are to add value and satisfy our customers. This is why suppliers are being treated as an extension of a company. The customer–supplier relationship is much closer than in the past, with confidential process information being exchanged if it promotes better quality. Partnerships are being formed over long-term periods, and customers are devoting time to “growing” their suppliers. As a result of the need to focus on the customer–supplier marriage, customers are vastly reducing the number of suppliers used, and the primary criterion for retaining them is their quality of products and services.

8.4. Deming's Thoughts on Continuous Improvement

One of the most outspoken proponents of continuous quality improvement is Dr. W. Edwards Deming. Dr. Deming has been a great supporter of the worker while being critical of management practices. While controversial, his teachings continue to grow in acceptance as more and more people realize that the management ways of the past have ceased to work in this worldwide competitive workplace. Dr. Deming's 14-Point Special Program may be paraphrased as in Figure 8.3.³

³This is simply an updating from the set of 14 points provided by Dr. Deming's office, Washington, D.C., Nov. 1983.

1. Create constancy of purpose toward improvement of product and service, with the aim to become competitive and to stay in business, and to provide jobs.
2. Adopt the new philosophy. We are in a new economic age. Western management must awaken to the challenge, must learn their responsibilities, and take on leadership for a change.
3. Cease dependence on inspection to achieve quality. Eliminate the need for inspection on a mass basis by building quality into the product in the first place.
4. End the practice of awarding business on the basis of price tag. Instead, minimize total cost. Move toward a single supplier for any one item, on a long-term relationship of loyalty and trust.
5. Improve constantly and forever the system of production and service, to improve quality and productivity, and thus constantly decrease costs.
6. Institute training on the job.
7. Institute leadership. The aim of supervision should be to help people and machines and gadgets to do a better job. Supervision of management is in need of overhaul, as well as supervision of production workers.
8. Drive out fear, so that everyone may work effectively for the company.
9. Break down barriers between departments. People in research, design, sales, and production must work as a team, to foresee problems of production and in use that may be encountered with the product or service.
10. Eliminate slogans, exhortations, and targets for the work force asking for zero defects and new levels of productivity. Such exhortations only create adversarial relationships, as the bulk of the causes of low quality and low productivity belong to the system and thus lie beyond the power of the work force.
11.
 - a. Eliminate work standards (quotas) on the factory floor. Substitute leadership.
 - b. Eliminate management by objective. Eliminate management by numbers, numerical goals. Substitute leadership.
12.
 - a. Remove barriers that rob the hourly worker of his right to pride of workmanship. The responsibility of supervisors must be changed from sheer numbers to quality.
 - b. Remove barriers that rob people in management and in engineering of their right to pride of workmanship. This means, *inter alia*, abolishment of the annual or merit rating and of management by objective.
13. Institute a vigorous program of education and self-improvement.
14. Put everybody in the company to work to accomplish the transformation. The transformation is everybody's job.

Fig. 8.3. Dr. Deming's elaboration on the 14 points. [From W. Edwards Deming, *Out of the Crisis* (Cambridge, Mass.: MIT Press, 1986), pp. 23–24.]

8.5. Juran's Contributions to Quality Thought

Dr. J. M. Juran has perhaps been the most prolific of the world's quality thought leaders. Dr. Juran, an engineer and lawyer, has leveraged his theories, observations, and knowledge to a worldwide audience through his writings and his consulting organization. Dr. Juran presents pragmatic, up-to-date, and detailed approaches for helping practitioners build an organization within which TQM is a reality.

Dr. Juran often speaks of his Juran Trilogy, which includes (1) quality planning, (2) quality control, and (3) quality improvement. *Quality planning* is determining current and future customer needs and developing products and processes to meet or exceed those needs. *Quality control* is evaluating actual performance, comparing it to goal, and taking action if the two are different. *Quality improvement* is the methodical attacking of waste (often thought of as non-value-added operations). In the remainder of this chapter we focus on some specific tools used in all three: planning, control, and improvement efforts.

8.6. Tools for On-Line vs. Off-Line Quality Control

We often hear people talk about off-line and on-line quality control. Figure 8.4 graphically presents a picture of off-line QC leading into on-line QC. *Off-line QC* refers to design of products and processes to produce them. Generally, these take place in advance, before the production system goes on-line. Off-line, there is a great deal of emphasis on correctly capturing the wants and needs of current and potential customers, then designing products and services to meet their needs, and then designing the necessary production processes. Experts agree that the majority of the deficiencies we realize during the production of goods and services is not because we are careless during the on-line phase, but because we actually do an insufficient job during the off-line phase. Some experts estimate that between 60 and 80% of the deficiencies we see are actually designed into the product or its production process. We can see that there is a great opportunity for the prevention of problems during the off-line phase.

On-line QC is usually synonymous with actual production activities. During production, our intent is to operate our production processes in a way that (1) meets targets, and (2) minimizes variation. That is, we want to operate our processes so stably, consistently, and predictably that our customers know that every time they buy from us, they will receive a consistent product or service. Stable processes are said to be "in control." Unstable processes are referred to as inconsistent, unpredictable, or out of control. If a process is stable and also meets customer requirements, it is said to be a capable process. We want all of our processes to be capable.

There are many tools that we can use during both off-line and on-line QC. Only some of the most successful and popular tools will be presented in the remainder of the chapter. These tools are shown in Figure 8.5. We discuss them in turn.

8.7. Quality Function Deployment

QFD is a concept that helps translate customer requirements into product and production requirements. The idea is to begin by learning and agreeing upon the customers' requirements, and to use that knowledge to drive development or improvement of the process. More time is spent up-front in the design phase, but that time is more than made up in time reductions in production since the product and processes were thought out well in the beginning. As the man says, "pay me now or pay me later." More up-front time spent on getting it right the first time is more than rewarded later as we make and sell product.

Figure 8.6 presents a QFD matrix that translates customer requirements (the voice of the customer) for a connector (plug or socket) into final product control (technical) characteristics. Another matrix might be used to translate the final product control characteristics into needed process characteristics. Still another matrix might be used to translate needed process characteristics into process control points.

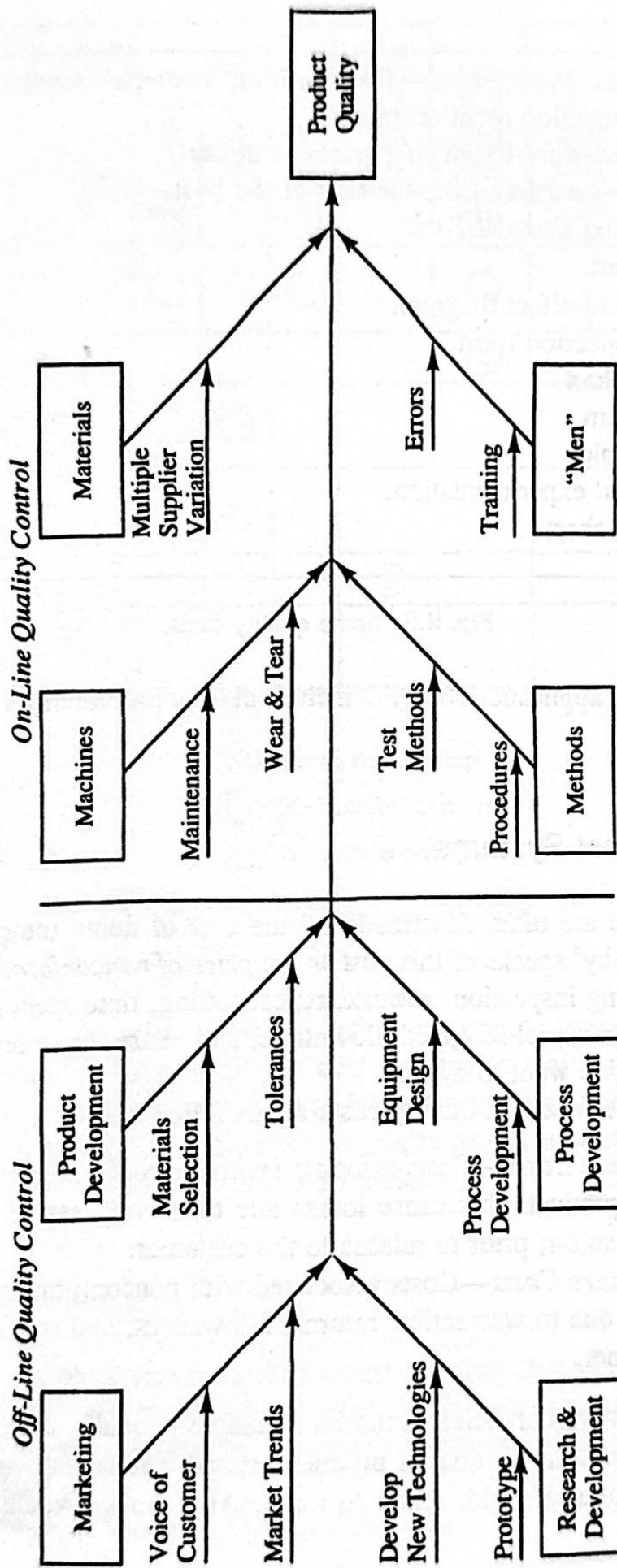


Fig. 8.4. On-line and off-line quality control.

Quality Function Deployment—Transforming customer requirements into product and production requirements.

Quality Cost Systems—Estimating waste in dollars.

Benchmarking—Learning from the best of the best.

Statistical Process Control Tools

- Flowchart.
- Cause-and-effect diagram.
- Data collection form.
- Pareto chart.
- Histogram.
- Scatter plot.
- Designed experimentation.
- Control chart.

Fig. 8.5. Some quality tools.

Typically, industry applications of QFD include at least four matrices. They do take time!

8.8. Quality Cost Systems

Quality costs are often described as “the cost of doing things wrong.” In fact, Mr. Phil Crosby⁴ speaks of this cost as the *price of nonconformance* (PONC). For example, sorting inspection, rework, repeat testing, time spent resolving customer complaints, material scrapped, downtime, and returns are a few good examples of the wastes we want to avoid.

Two major elements of quality costs are as follows:

- (1) **Internal Failure Costs**—Costs associated with nonconforming materials, components, or products that cause losses due to rework, repair, retest, scrap, sorting, and so on, prior to release to the customer.
- (2) **External Failure Costs**—Costs associated with nonconforming products that cause losses due to warranties, returns, allowances, and so on, after release to the customer.

A quality cost system helps us monitor changes in quality costs: for example, a recent trend upward in the cost of product returns. The quality cost system also helps us identify pots of gold, ready to mine. An example would be in-process

⁴Philip Crosby and his firm have provided quality awareness and training for thousands of practitioners.

Technical Characteristics Customer Needs		Ends			Number of Components	End Finish Technique	Pull-out Force	Roundness	Crimping Method
		ECC	Mar'l	Float					
Good Performance	Low Loss	⊙		⊙		⊙		○	
	Repeatable Loss	○	△	⊙		○			
	Low Reflection					⊙		⊙	
	Wide Temp. Range		○						
Low Cost	Inexpensive Materials								△
	Fast Assembly				⊙				
	Inexpensive Tools								○

- ⊙ Strong Relationship
 ○ Medium Relationship
 △ Weak Relationship

Fig. 8.6. QFD matrix-connector.

inventories that are consistently high, eating up capital, space, and morale. The important concept underlying quality cost systems is that they identify areas to prioritize for process improvement. Such investments are usually highly leveraged, yielding a 6:1 return. That is, if we are willing to pay \$1 now to prevent problems, we will save \$6 in the absence of those problems.

8.9. Benchmarking

Benchmarking is the search for industry best practices that lead to superior performance. It has also been described as the continuous process of measuring products, services, and practices against the toughest competitors or those companies recognized as industry leaders.

The four main features of benchmarking are:

- (1) We are striving to learn from the "best of the best."
- (2) We are striving to be the "best of the best."

- (3) We are *externally oriented*, looking outside our company and even our industry.
- (4) We are focusing first on best *practices* (how to do key work processes) and then on *metrics* or measures (to assess performance).

As examples, suppose that you are a specialty electronics manufacturer. You want to benchmark warehouse material handling. You might go completely outside your industry, to a mail-order dry goods business that is well known for their warehouse material handling efficiency. Or, suppose that you have a critical, expensive, and much needed set of precision painting areas that have specialized capabilities. Scheduling them is a fright. You might wish to benchmark a world-renown hospital to determine how they schedule operating rooms.

Example 8.1

As president of Gadgets, Inc., you and your employees are determined to produce a high-quality product. One of your teams has been through a highly successful QFD effort and you believe Gadgets now has a solid fix on customer requirements, product characteristics, and processes necessary to efficiently and effectively deliver the product—the new supergadget. Simultaneously, recent quality cost figures have been assessed and you have determined that there is at least one begging need—for improvement in the consistency of raw materials received from suppliers. Variations in their product cause all sorts of instability and constant readjustment in Gadgets' machining operations. You have benchmarked an outstanding firm known for excellent relations and results with suppliers. You have determined many of the practices they have used, applying the tools of statistical process control to assist their suppliers.

You have also implemented many of Dr. Deming's 14 points. For example, you are no longer using end-of-line inspection to try to inspect quality into the product. Rather, you are focusing on your processes upstream in the process to ensure that quality is built in from the beginning. You no longer submit requests for bid to any and all potential suppliers, awarding business on the basis of first cost alone. Rather, you have selected your overall best supplier for each type of raw material, and have decided to "grow" those suppliers as if they were part of your own organization. That is, you plan to nurture those suppliers as if they were an extension of Gadgets, Inc., embracing them in a long-term relationship.

You are focusing on both product and service delivery processes and have implemented rigorous training of all employees. Management and supervisors are now thrilled when employees surface problems to be worked, rather than hiding those problems out of fear. Different parts of the company are working together as a team. For example, purchasing, materials management, design engineering, process engineering, and marketing plan to work together on the team to help improve the key suppliers. Some members of the team will be hourly workers; others will be salaried personnel.

Even though reducing the supplier base got rid of a lot of variation and quality problems, there is still room for improvement. You think there is great potential to work closely with suppliers, using the tools of SPC, to help them become a better supplier to you. First, however, it is time to review those tools of SPC.

8.10. Tools of Statistical Process Control⁵

The tools of SPC are absolutely necessary to help us understand and improve processes. These tools help teams communicate, share and document ideas, understand variation, and measure results of process changes. The eight common tools of SPC are pictured in Figure 8.7. Seven of them will be addressed briefly in this section. Control charts will be treated separately and more extensively.

8.10.1 Flowchart

A *flowchart* is a picture of the activities that take place in a process. Drawing a flowchart is usually one of the first things a team does. It gives everyone on the team a chance to communicate about, understand, and agree on how the process is accomplished. It defines customer-supplier relationships and helps standardize procedures. It is a simple but powerful tool. An invoice-processing flowchart is illustrated in Figure 8.8.

Several guidelines are helpful to us in drawing a flowchart:

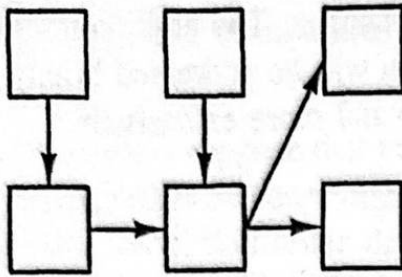
- (1) Establish boundaries for the process you want to flowchart. Exactly where does it start and where does it stop?
- (2) Try to hit a level that shows the truly key steps in the process, without becoming too detailed. If the level is too shallow, you can go back in and draw in more detail. If the level is too deep, you may get bogged down in detail that is not truly helpful.
- (3) Keep the symbols simple. There is little to be gained by using all of the common "computer program" flowchart symbols.
- (4) Be sure that there are no dead ends (other than at "stop") or loops that continue forever.
- (5) Show those feedback loops where something goes back for rework, checking, and so on.

After drawing the flowchart, keep a few things in mind:

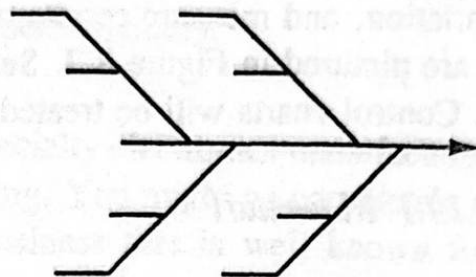
- (6) After drawing a chart of the existing process, we might want to "imagineer" a flowchart of the process if everything worked right. Then the two can be compared and important areas for improvement may be identified.
- (7) Feedback loops are a likely sign of waste and opportunity for improvement.
- (8) Consider collecting data immediately following a "decision" diamond. This is where you can gain some real insight as to what is really going on in the process.
- (9) Between every pair of "action" or "step" boxes is a customer-supplier relationship.

⁵Much of this material is taken directly from the Exxon Chemical Company Quality Principles Course manual.

1. Flow Chart



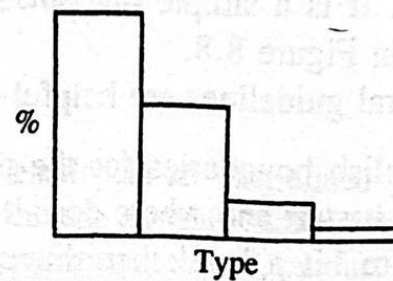
2. Cause and Effect



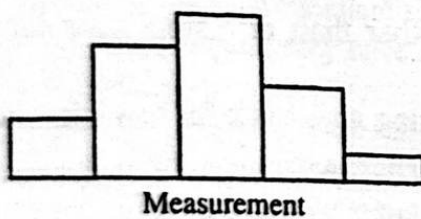
3. Check Sheet

A	B	C	D	E
✓	✓	✓		✓
✓		✓		✓
✓		✓		
✓		✓		
✓		✓		

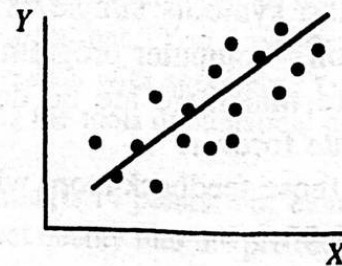
4. Pareto Analysis



5. Histogram



6. Scatter Plot



7. Designed Experimentation

	A	B	C	D
I	X	X	X	X
	X	X	X	X
II	X	X	X	X
	X	X	X	X

8. Control Chart

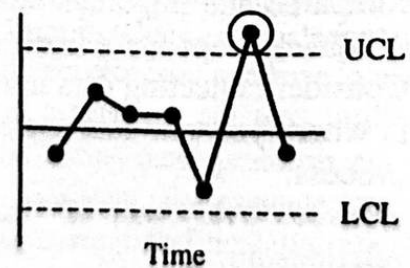


Fig. 8.7. Eight basic SPC tools.

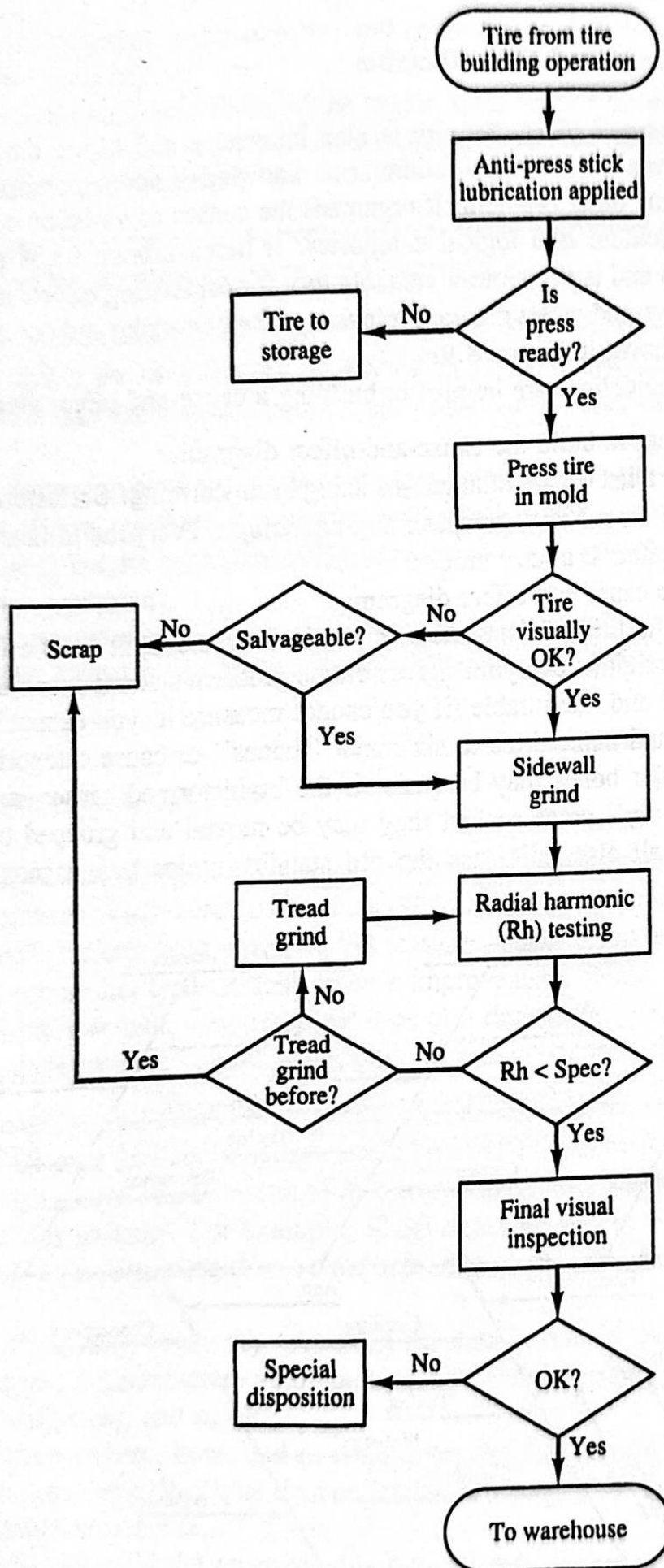


Fig. 8.8. Tire cure and finish process flowchart.

8.10.2 Cause-and-Effect Diagram

The *cause-and-effect diagram* is also known as a *fishbone diagram* or an *Ishikawa diagram*. It is used to summarize knowledge about possible causes of variation or some other problem. It organizes the causes of variation or the causes of a quality problem into logical categories. It helps a team focus on different possible causes and is therefore a valuable tool for organizing efforts to improve a process. A cause-and-effect diagram relevant to the ham radio station of K5KC and WA5DSH is shown in Figure 8.9.

Several guidelines are helpful in building a cause-and-effect diagram:

- (1) Use a team to build the cause-and-effect diagram.
- (2) Generate a list of potential causes using brainstorming. Brainstorming allows everyone on the team to speak and encourages everyone to listen. All ideas are evaluated at a later time.
- (3) Build the cause-and-effect diagram:
 - Place the problem statement in the box on the right (at the fish's head). Make sure that everyone agrees on the problem statement and that it is well defined and measurable (if you cannot measure it, you cannot improve it).
 - Draw and name three to six major "bones" or cause categories. Naming the major bones may be easier if the brainstormed causes are written on "stick-um" notes so that they may be moved and grouped by the team. When all else fails, use the old standby major bones: men, machines, materials, methods, money, and measurement.

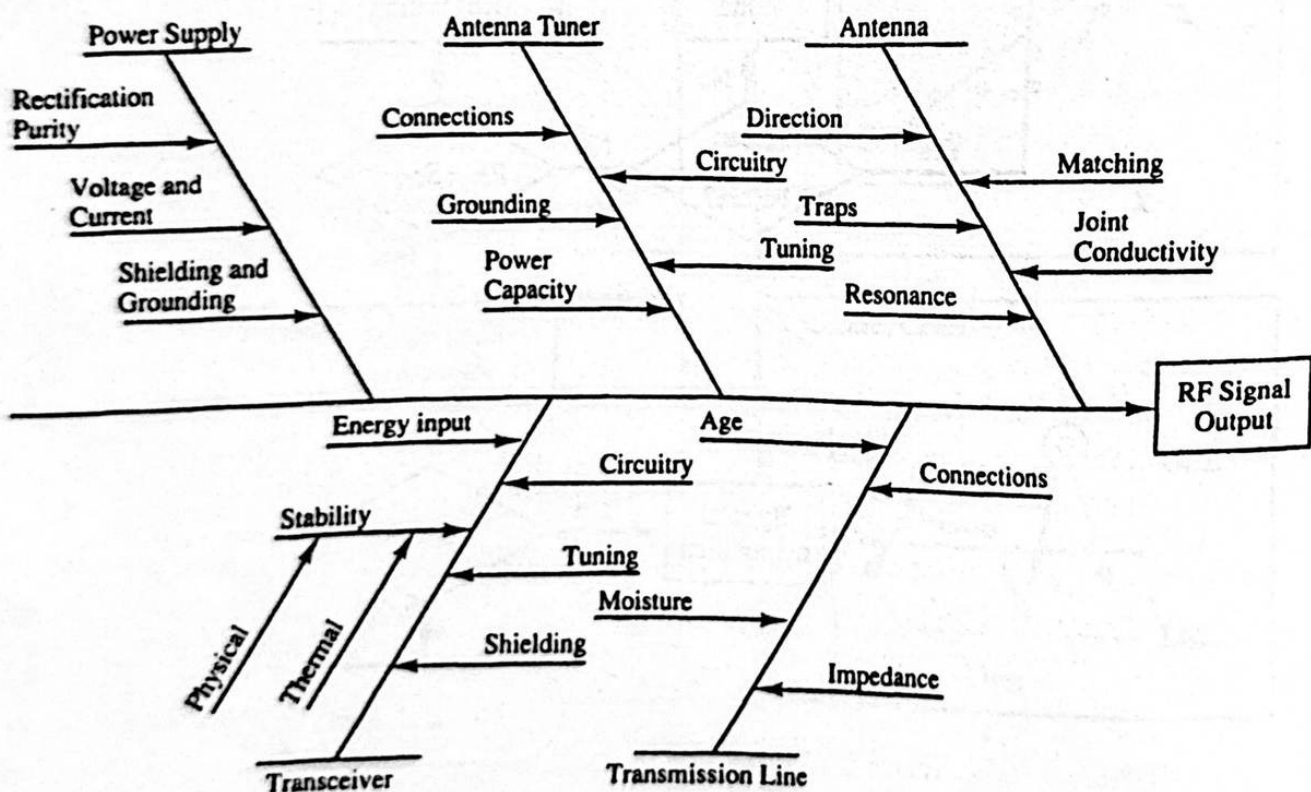


Fig. 8.9. Cause-and-effect diagram for ham stations K5KC and WA5DSH.

methods, materials, measurement, and environment, but these are seldom the best breakouts.

- Place the brainstormed ideas on the major, most related bones.
- For each cause, ask "Why does it happen?" and list the responses as subbones or sub-subbones.

After drawing the cause-and-effect diagram, focus on finding the most important causes of the problem by doing the following:

- (4) Look for causes that appear repeatedly. Data may be needed to identify these.
- (5) Discuss each of the causes listed, as desired by the team. Use "expert knowledge" from the team, plus the question "why" to identify the most basic causes.
- (6) Reach a team consensus as to which of the possible causes deserves the most attention. Focus on these for greater understanding and knowledge of the process, with the intent of eliminating or reducing those causes that are clearly affecting the problem.
- (7) Keep the cause-and-effect diagram updated as knowledge of the process improves. It is a working document that should capture the "brains" of the team on paper.

8.10.3 Data Collection Form

Part of any effort to improve a process is to gain knowledge about the process. This usually requires us to collect and analyze data. Unfortunately, most of the data we *routinely* collect alerts us to problems but is useless for analysis and helping us take the right action that leads to real process improvement. When we decide to define and collect new data, we need some type of a *data collection form*.

Several guidelines are helpful in designing a data collection form:

- (1) Decide exactly what data is to be collected. For each characteristic, state its units of measure (dollars of scrap/month; hours overtime/week, etc.)
- (2) Clearly define each characteristic so that everyone has a common understanding of it and its units. For example, if the characteristic is "time lost" in hours per week, there should be a clear definition of exactly what "time lost" means.
- (3) Decide if you wish to stratify (separate) the data according to some factor. For example, different strata might be by plant, shift, continent of customer, type of equipment, and so on.
- (4) Decide when, where, how, and in what order the data should be collected. The idea is to "visualize" the data collection process and to make it as easy and failproof as possible.
- (5) Determine and build the most suitable type of data collection form. Three types receive a lot of use.

- First, a *traditional data collection form* may be designed. It should have a clear place for all measures to be recorded, and its layout should conform to the order in which data are collected.
- Or a simple *checksheet* might be designed to collect data in the form of "check marks." The categories into which check marks are placed should be well defined so that there is no overlap between them. A defect identification check sheet is shown in Figure 8.10 for bales of butyl rubber just produced.

Some hints on using the data collection form include:

- (6) After designing and building a data collection form, give it a trial under fire. Let those who will be collecting the data try it for a day or two. Be sure that they are people who have *not* been involved in the design of the form.
- (7) Take their comments and make design changes as needed.
- (8) Nearly all data must be converted into information. Plotting the data on a Pareto chart, frequency plot, scatter plot, control chart, or other device are common methods of making sense out of data.

8.10.4 Pareto Analysis

"Pareto" is "prioritization." A *Pareto analysis* orders occurrence data by type, category, or other classification. It helps us focus on the important things. It identifies the "vital few" problems or types of defects from the "useful many." This helps us pinpoint the right problem or effect to study. The "Pareto principle," also known as the "80/20 rule," says, in effect, that 80% of our "heartburn" is caused by 20% of the things with which we deal. A Pareto diagram showing reasons for a chemical unit's downtime is shown in Figure 8.11.

Several guidelines are helpful in building a Pareto diagram and doing Pareto analysis:

- (1) Decide the data that are to be collected. This includes deciding on the measurement axis of the diagram. It is usually a measure of "pain," such as dollars per year, number of occurrences per month, downtime per week, or number errors per 100 invoices. It includes deciding on the classification axis of the diagram. This is usually a type, category, or classification of defect. Examples include type of injury, type of defect, area of plant, day of week, country from which order placed, and so on.
- (2) Clearly define each classification that will be along the classification axis. These should cover all possible classifications and not overlap in definition.
- (3) Design a data collection form. Begin to collect data. After collecting at least 30 occurrences, set up the Pareto diagram and look for the most frequently occurring classification. Pareto diagrams order the classifications on the axis from most frequently occurring to least frequently occurring.

Date 12/14/9x Shift AM Line #2

REASON FOR INSP. (a)	0600 - 0800					0800 - 1000					1000 - 1200					1200 - 1400					1400 - 1600					1600 - 1800				
	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E					
	BALERS																													
WEARSTRIP WORMS (Blacks, stringy, soft piece of rubber)																														
BLACK SPECKS (Unidenti- fiable, soft black spot)																														
GREASE (Black or greenish spot, may rub off on hand or may be mixed into rubber)																														
RUST (Reddish hard magnetic)																														
COLOR (Noticable change in rubber color, usually caused by poly upset)																														

Fig. 8.10. Check sheet for defects in butyl rubber.

Note 1: Lack of feed totaled 98.5 hrs

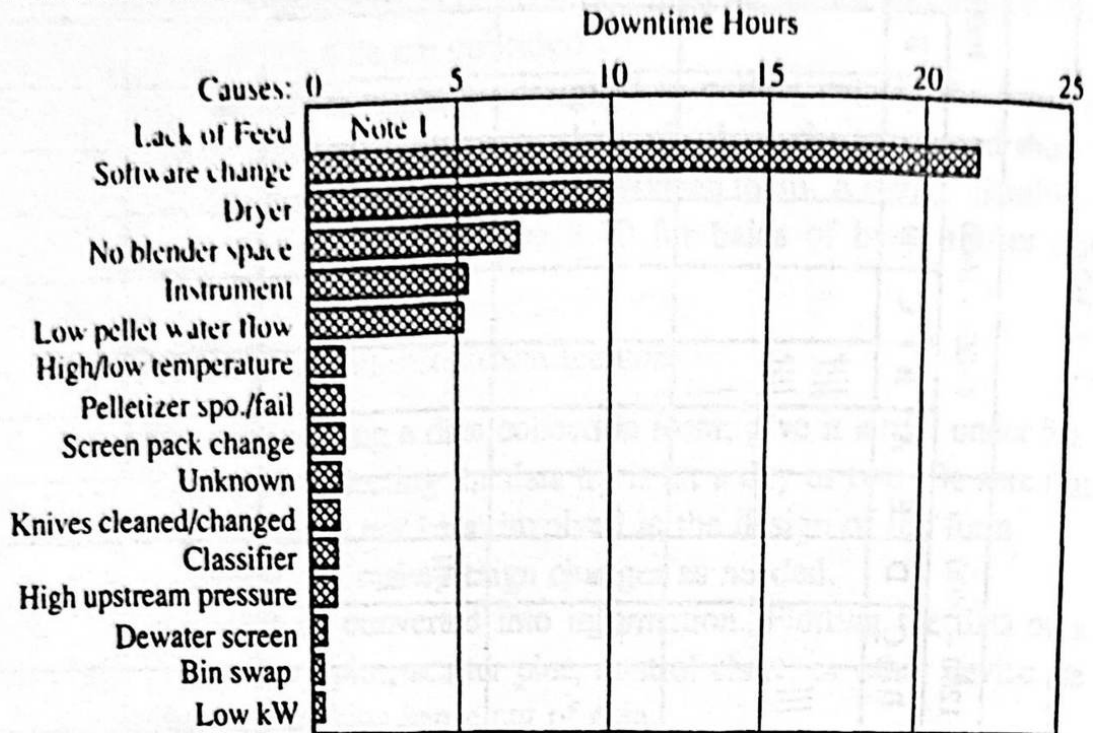


Fig. 8.11. Pareto chart of downtime causes.

Some helpful hints in using Pareto analysis include:

- (4) Be sure that the measurement axis is really a measure of "pain." If there are multiple measures that are equally meaningful (e.g., percent downtime and number of occurrences), do a Pareto diagram for each.
- (5) A Pareto diagram may be broken down further. That is, Pareto analysis may be done on data from the biggest bar of the Pareto diagram. Repeating this several times often leads to the right aspects of the process to study. This is often called "macro-to-micro" or "Pareto within Pareto" analysis. For example, data from the most frequent invoice error may again be broken down by type of invoice.
- (6) Spend the most time on defining the classifications along the classification axis. These should be crystal clear, or the data will be misleading due to improper classification.

8.10.5 Histogram

The histogram presents a snapshot in time of a set of data. It is particularly useful for seeing the shape, centering, and spread of a set of data from some process. If the process is stable (see control charts later in this chapter), the histogram is a picture of what we might expect to see in the future. If the process is not stable, the histogram is only a picture of what has happened in the past. A histogram of student test scores is presented in Figure 8.12.

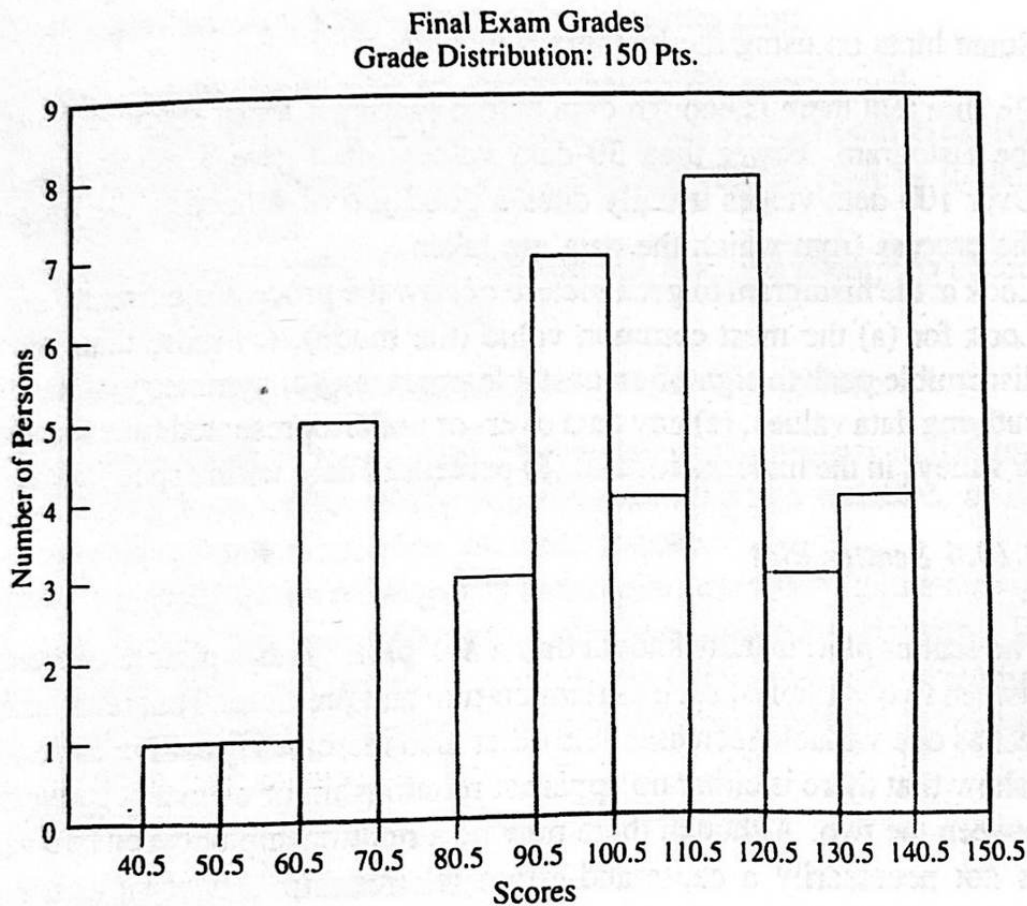


Fig. 8.12. Histogram of the test scores.

Several guidelines are helpful in building a histogram:

- (1) The characteristic of interest (e.g., height, weight, viscosity, melt point, time, etc.) is scaled on the horizontal axis.
- (2) The scale of the characteristic of interest is usually broken into cells of equal width.
- (3) It is best if cell boundaries are such that data must fall inside a cell and cannot fall on a cell boundary. It is common to set up cell boundaries to one decimal place past the data, with the last place ending in a 5. For example, cell boundaries of 7.5 to 11.5 would include data numbered 8, 9, 10, and 11.
- (4) The number of cells is often from 5 to 20, with 10 being a good starting number. To get the cell width, begin by taking the difference between the maximum and minimum number and dividing that difference by the approximate number of histogram cells desired. Then round that to a workable cell width. For example, if the largest and smallest data values are 77 and 53, respectively, and about 10 cells are desired, $(77 - 53)/10 = 2.4$. Use cells of either width 2 or 3.
- (5) Either the frequency or the percent of occurrences is scaled on the vertical axis.

Some hints on using the histogram include:

- (6) Be sure that there is enough data before putting a lot of stock in the shape of the histogram. Fewer than 30 data values often give a misleading picture. Over 100 data values usually does a good job of reflecting the operation of the process from which the data are taken.
- (7) Look at the histogram to get a picture of how the process is currently operating. Look for (a) the most common value (the mode), (b) more than one clearly discernible peak (a sign of an unstable process), (c) symmetry of the data, (d) outlying data values, (e) any data over- or underrepresented (unexpected peaks or valleys in the histogram), and (f) percent of data within specification limits.

8.10.6 Scatter Plot

The scatter plot, usually shown on an X-Y plot, gives a picture of the relationship between two variables, such as temperature and pressure. That relationship may show that as one variable increases, the other also increases (positive correlation); or it may show that there is either no apparent relationship or even a negative relationship between the two. Although there may be a relationship between two variables, there is not necessarily a cause-and-effect relationship between them. A scatter diagram showing business calls answered versus calls received (some persons hung up before someone answered the phone—all call data recorded electronically) is shown in Figure 8.13.

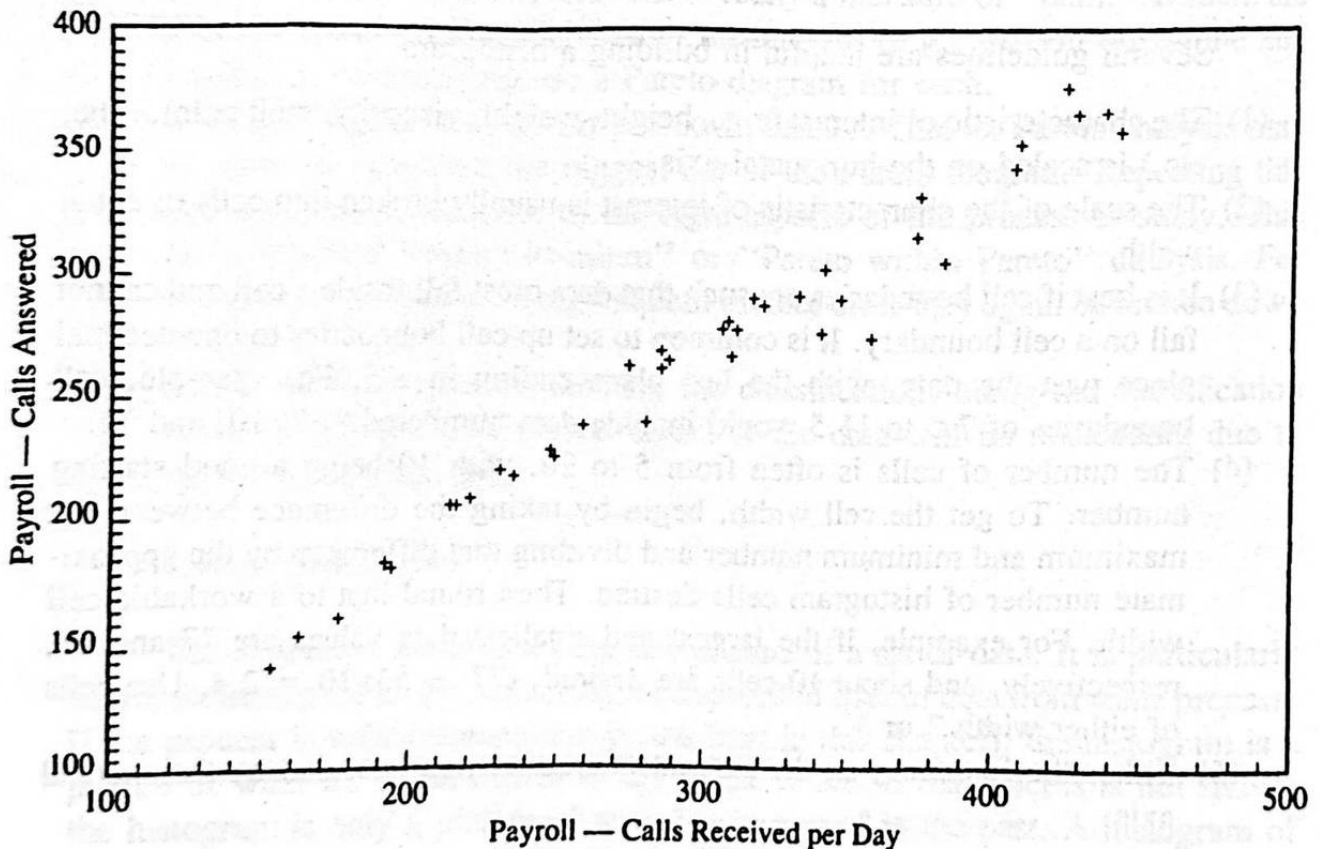


Fig. 8.13. Scatter plot of calls answered or received.

Several guidelines are helpful in building a scatter plot:

- (1) Draw two perpendicular axes of approximately the same length.
- (2) Scale the axes so that the data collected use about 90% of both axes. Be sure that both scales are increasing as you go up the vertical scale or to the right on the horizontal scale.
- (3) Plot the data pairs on the *X-Y* plot, look at the data, and attempt to understand the process and how it is operating.

Some hints for using and interpreting the scatter plot include:

- (4) If there is a relationship, use your knowledge of the process to determine if there is a cause-and-effect relationship between the two variables, or if there is some other common variable influencing both.
- (5) If there is a straight-line relationship in the plot, eyeball-fit a line through the data (this can be done more formally using least-squares regression). Do not extend the line beyond the data. That is, do not extrapolate, assuming the process will continue to operate in a straight-line relationship well beyond the data collected.

8.10.7 Designed Experimentation

Designed experiments are one of the most powerful tools we have in our toolkit to improve processes. Basically, we are attempting to determine economically which of many factors have the most significant effect on process centering and spread. We would like not only to be on target, but also to minimize variation.

Everyone has done designed experiments. For example, in high school or in a college science lab, you probably varied a factor such as temperature or pressure or slope or time to see its effect. Typically, such experiments are 1FAT designs (one factor at a time). They are particularly inefficient at best, and misleading at worst. They are not nearly as powerful as if multiple factors are varied simultaneously in a particular pattern as designated by one of many possible experimental designs. Common designs used effectively in industry include factorials, fractional factorials, Taguchi designs, Plackett–Burman screening designs, central composite designs, and others.

Experimental designs can help us study quantitative effects (e.g., temperatures of 100 and 140 degrees) and qualitative effects (e.g., machine A vs. machine B). We design the experiment in advance to ensure that we will be able to discern if the effects of interest are important factors in our process. Proper designs let us test for interaction effects. An interaction exists, for example, if process yield is better at high temperature when raw material A is used, but yield is better at low temperature if raw material B is used.

Designed experimentation is one of the most needed and important bodies of knowledge you can have today in industry. If you do not get a course in this in industrial engineering, take it as an elective in the statistics department. Tell your

other engineering friends (non-I.E.s) that they need it, too. An example of data and marginal mean plots is shown in Figure 8.14.

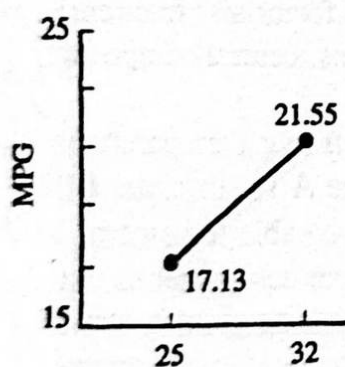
In this section we have had a very practical introduction to many of the SPC tools. Next we cover one of the most important and useful tools, control charts, in more depth.

8.11. Background on Control Charts

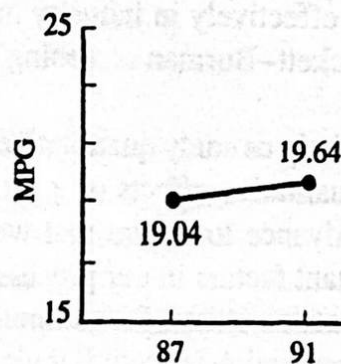
Control charts are a tool for studying variability. They tell us if a process is showing stable (or consistent) variation. A stable process is often called an in-control process, a predictable process, or a "common cause" process. It is said to be in a state of statistical control (SOSC). An unstable process is also known as out-of-control (OOC), unpredictable, or a "common plus special cause" process. A control chart tells us whether or not a process is stable.

Everything we do in life is a process. Everything in life varies. Brushing your teeth is a process. You have never brushed your teeth twice in exactly the same way. We want to learn about variation and how it affects our work processes. More specifically, we are trying to determine whether or not all of our variation is due to

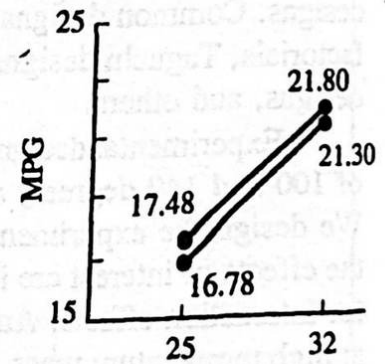
Tire Pressure	Gasoline Octane	Miles per Gallon Trial				Mean $\bar{y}$
		y_1	y_2	y_3	y_4	
25	87	16.5	18.3	15.7	16.6	16.78
25	91	18.2	16.6	17.3	17.8	17.48
32	87	21.3	22.6	19.5	21.8	21.30
32	91	21.5	20.9	23.1	21.7	21.80



Tire Pressure
Effect: > 4 MPG



Gasoline Octane
Effect: < 1 MPG



Tire Pressure
Interaction: None
(Since $\approx$ Parallel Lines)

Fig. 8.14. Designed experimentation on car gasoline mileage.

"common causes" or if "special causes" are also present. All processes have variation due to common causes.

Common causes are those that are inherent in the process over time, affect everyone working in the process, and affect all outcomes of the process. Learning about industrial engineering is a process. Common causes to all students in an introduction to industrial engineering course include (1) your textbook, (2) the classroom, (3) the lighting in the room, (4) the teacher, (5) the number of credit hours, and so on.

Special causes are those that are not part of the process all the time, do not affect everyone, but arise due to specific circumstances. Examples would be an individual's mathematics background, their experience working in industry, whether or not they are sick, and so on. If special causes are at work, they can be eliminated by attacking their underlying specific causes (e.g., remedial math, summer industry experience, medical treatment followed by preventive medicine). If common causes are at work, the only way to reduce or remove them is to make fundamental changes in the system or the process (e.g., change classrooms, improve lighting, modify the number of credit hours, etc.). *Note:* This book and your instructor probably cannot be improved!

Control charts have been used as a diagnostic and maintenance tool in the control of production processes since first introduced by Shewhart, the father of control charts. By far the most popular control charts are the $\bar{X}$, R , p , and c charts. The $\bar{X}$ and R charts are used together on variables data (e.g., numerical measurements) to analyze the central tendency and spread of a process on a single measurable characteristic. The p chart is used to analyze attributes data (e.g., good/bad classification), such as the fraction nonconforming of a process. The c chart is used when attributes data, such as a count of the number of nonconformities per sample unit, is the measure of interest.

Control charts can be used as a management tool to help (1) bring a process into control, (2) keep a process in control, and (3) determine process capability to meet specifications. Output from a process is sampled and a statistic is calculated which bears the name of the control chart on which it is to be plotted (e.g., the sample average $\bar{X}$ is plotted on the $\bar{X}$ chart). When process variability is due only to *common* causes, the process is in a *state of statistical control* (SOSC) and the plotted pattern of statistics over time is reasonably well behaved, falling within predictable limits and according to predictable patterns. When process variability is due to *common and special* causes, such as people, equipment, and/or materials, the process is *out of control* (OOC) and nonrandom patterns appear on the control chart.

Control charts have a center line plus upper and lower control limits. These control limits are usually spaced 3 standard deviations (of the sample statistic being plotted) above and below the center line. As such, it is highly unlikely that a plotted point will fall outside control limits as long as an SOSC exists. In addition to points outside control limits, the *pattern* of plotted points within control limits is equally

important in identifying an OOC condition and its cause. Therefore, sample statistics are almost always plotted in time order.

8.12. Control Charts for Variables

The $\bar{X}$ and R control charts are used together in analyzing a single measurable characteristic. Approximately $m = 20$ or 30 subgroups of size n each are selected. Typical subgroup sizes are $n = 4$ or 5, selected consecutively from the process or produced under as nearly identical conditions as possible. Time between subgroups is dependent on judgment and may be once per hour, twice per day, once per shift, and so on. The idea is to have *within*-subgroup variation be as small as possible, and represent common-cause variation only. Production process differences, if any, then showing up over time *between* subgroups, and represent special cause variation. For each subgroup i , the average $\bar{X}_i$ and range R_i statistics are calculated.

Once the desired number of subgroups m has been inspected, the next step is to calculate the average range, $\bar{R}$, of the subgroups. This will be the center line on the R chart. Factors for the upper and lower control limits are given as D_4 and D_3 , respectively, in Table 8.1. The equations pertinent to an R chart based on m subgroups are given as follows:

$$\text{center line} = \bar{R} = \frac{\sum_{i=1}^m R_i}{m} \quad (8.1)$$

$$\text{upper control limit} = \text{UCL}_R = D_4 \bar{R} \quad (8.2)$$

$$\text{lower control limit} = \text{LCL}_R = D_3 \bar{R} \quad (8.3)$$

Note that for small subgroup sizes, three standard deviations below $\bar{R}$ would yield a negative range control limit, which is impossible. Therefore, when the subgroup size is six or less, D_3 is set to zero, resulting in $\text{LCL}_R = 0$.

The $\bar{X}$ chart control limits depend on $\bar{R}$; therefore, it is important to establish the R chart first. The $\bar{X}$ chart assumes an underlying normal distribution in accordance with the central limit theorem. Also, on the $\bar{X}$ chart we are dealing with a distribution of subgroup averages with variance

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} \quad (8.4)$$

Our control limits will be set at a point equivalent to three subgroup average standard deviations above and below $\bar{X}$. The equations for the $\bar{X}$ chart, assuming m subgroups, are given as follows:

$$\text{center line} = \bar{\bar{X}} = \frac{\sum_{i=1}^m \bar{X}_i}{m} \quad (8.5)$$

Table 8.1. CONTROL CHART FACTORS

Subgroup Size, n	LCL_R Factor, D_3	UCL_R Factor, D_4	$CL_{\bar{X}}$ Factor, A_2	$\bar{R}/\sigma$ Ratio, d_2
2	0	3.267	1.880	1.128
3	0	2.575	1.023	1.693
4	0	2.282	0.729	2.059
5	0	2.115	0.577	2.326
6	0	2.004	0.483	2.534
7	0.076	1.924	0.419	2.704
8	0.136	1.864	0.373	2.847
9	0.184	1.816	0.337	2.970
10	0.223	1.777	0.308	3.078
11	0.256	1.744	0.285	3.173
12	0.284	1.716	0.266	3.258
13	0.308	1.692	0.249	3.336
14	0.329	1.671	0.235	3.407
15	0.348	1.652	0.223	3.472
16	0.364	1.636	0.212	3.532

$$\text{upper control limit} = UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R} \quad (8.6)$$

$$\text{lower control limit} = LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R} \quad (8.7)$$

The factor A_2 , values of which are given in Table 8.1, takes account of the relation between the range R , the process variance σ^2 , the subgroup average variance $\sigma_{\bar{X}}^2$, and our desire to have $\pm 3\sigma_{\bar{X}}$ control limits based on only common-cause or within-subgroup variation. For subtle reasons beyond the scope of this text, it is important that the control limits for both the $\bar{X}$ and R charts be calculated using the formulas given above rather than trying to take some alternative approach to the calculation of control limits.

Example 8.2

You have now embarked on a course of assisting one of your suppliers. Your quality engineer (Q.E.), while at the vendor's location, collects 30 subgroups of size $n = 4$ over several days, testing and recording the strengths of each unit of raw material. The data, averages, and ranges are given in Table 8.2. The center lines and control limits are calculated as follows:

$$\begin{aligned} \bar{R} &= \frac{\sum_{i=1}^m R_i}{m} = \frac{6,445}{30} = 214.83 \\ UCL_R &= D_4 \bar{R} = 2.282(214.83) = 490.24 \\ LCL_R &= D_3 \bar{R} = 0.0 \end{aligned}$$

Table 8.2. STRENGTH TEST DATA OF RAW MATERIAL

Subgroup	Sample Data				Subgroup Average, $\bar{X}$	Subgroup Range, R
	X_1	X_2	X_3	X_4		
1	908	1,095	1,159	1,215	1,094.25	307
2	1,337	1,080	1,086	1,242	1,186.25	257
3	1,212	1,326	1,175	1,380	1,273.25	205
4	1,286	1,278	1,151	1,359	1,268.50	208
5	1,401	1,313	1,384	1,466	1,391.00	153
6	1,114	1,096	1,060	1,204	1,118.50	144
7	1,119	974	1,007	1,212	1,078.00	238
8	1,001	1,030	1,063	1,037	1,032.75	62
9	1,152	1,161	1,159	1,046	1,129.50	115
10	1,000	882	870	823	893.75	177
11	835	667	787	803	773.00	168
12	707	904	961	918	872.50	254
13	949	809	842	859	864.75	140
14	828	1,061	918	964	942.75	233
15	1,043	753	813	861	867.50	290
16	784	847	856	945	858.00	161
17	1,086	925	855	840	926.50	246
18	1,117	1,201	1,034	929	1,070.25	272
19	1,115	1,114	1,038	1,227	1,123.50	189
20	1,140	928	1,332	1,061	1,115.25	404
21	1,229	1,127	1,030	1,253	1,159.75	223
22	1,314	1,242	1,087	1,140	1,195.75	227
23	1,305	1,219	1,353	1,207	1,271.00	146
24	1,181	1,326	1,245	1,548	1,325.00	367
25	1,312	1,285	1,444	1,176	1,304.25	268
26	993	1,052	1,018	1,074	1,034.25	81
27	1,196	980	1,116	1,161	1,113.25	216
28	948	1,139	1,222	1,062	1,092.75	274
29	993	1,283	1,231	1,105	1,153.00	290
30	1,050	928	1,009	1,058	1,011.25	130
					32,540.00	6,445.00

$$\bar{\bar{X}} = \sum_{i=1}^m \frac{\bar{X}_i}{m} = \frac{32,540}{30} = 1084.67$$

$$UCL_x = \bar{\bar{X}} + A_2 \bar{R} = 1,084.67 + 0.729(214.83) = 1,241.28$$

$$LCL_x = \bar{\bar{X}} - A_2 \bar{R} = 1,084.67 - 0.729(214.83) = 928.06$$

The control charts for the data given in Table 8.2 are drawn in Figures 8.15 and 8.16.

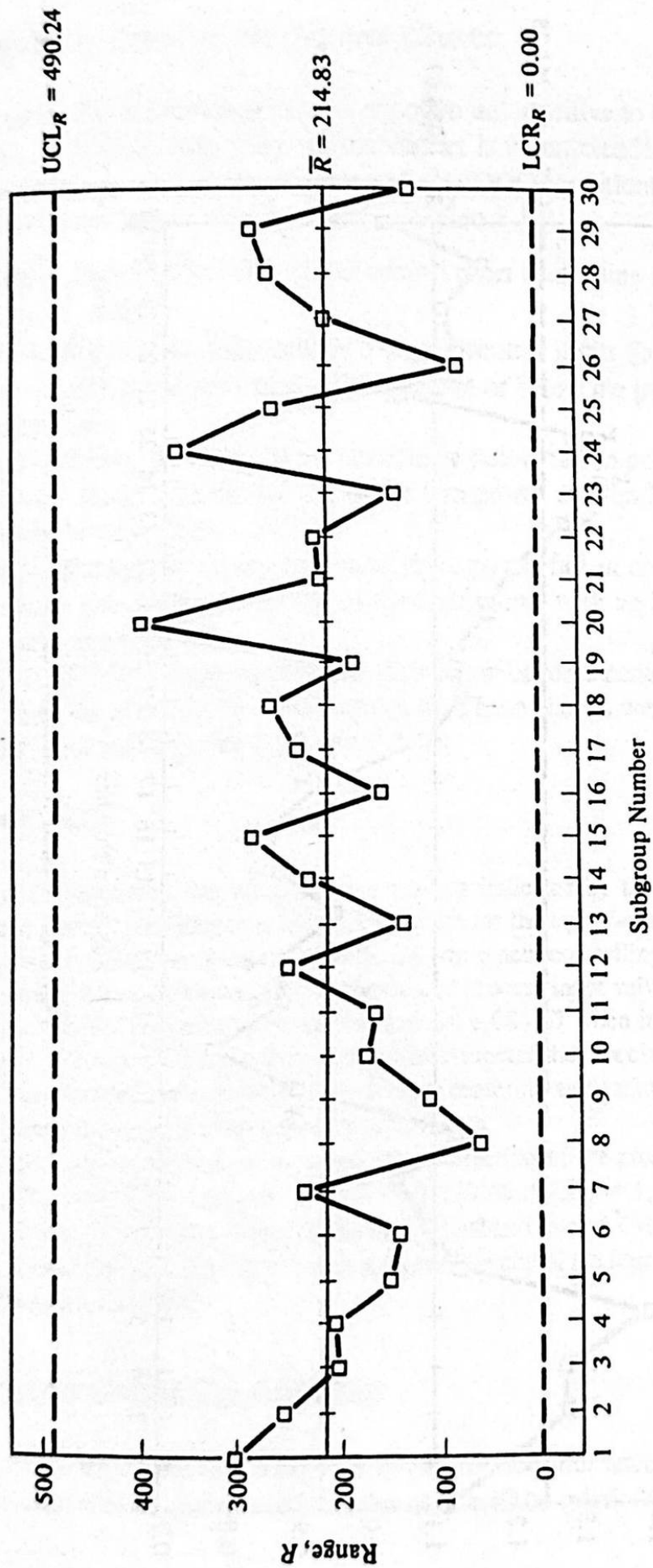


Fig. 8.15. *R* chart for strength test data.

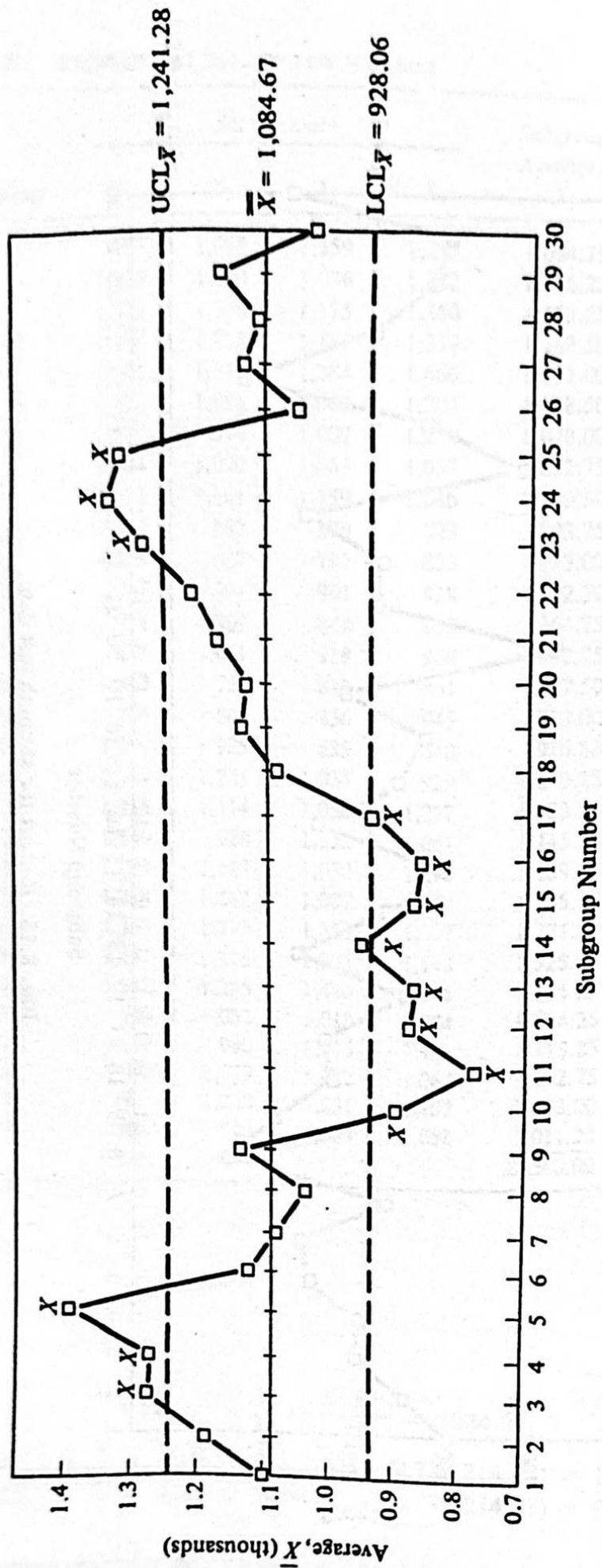


Fig. 8.16. $\bar{X}$ chart for strength test data.

8.13. Sensitivity Checks for Control Charts

When used alone, Shewhart control charts are often not sensitive to relatively small process shifts. A scheme consisting of four checks is recommended for making a control chart ($\bar{X}$, R , p , etc.) much more sensitive to OOC conditions.⁶ The steps of the scheme are listed below and illustrated in Figure 8.17.

- (1) Divide the area on either side of the control chart center line into three equal zones: A, B, and C.
- (2) *Check 1*—If one point falls outside 3-sigma control limits (beyond zone A), mark each such point with an X slightly above or below the point, away from the center line.
- (3) *Check 2*—If two out of any three successive points fall in or beyond zone A of the same side, mark the second of the two points with an X. The “other” point may be anywhere.
- (4) *Check 3*—If four out of any five successive points fall in or beyond zone B of the same side, mark the fourth of the four points with an X. The “other” point may be anywhere.
- (5) *Check 4*—If eight successive points fall in or beyond zone C of the same side, mark the eighth such point with an X. These checks were applied to the $\bar{X}$ and R charts of Figures 8.15 and 8.16.

Example 8.3

Your Q.E. concluded that the process spread as indicated by the R chart was in a SOSC, but that process centering was OOC. The cause for the out-of-control state was due to a faulty thermal isolation network on the metering equipment controlling the process. This was not the only problem, however. One of the critical process input valves was incorrectly set, causing the process centering to be far too low (at 1,084.67 when it should have been approximately 1,232.60). Correcting these difficulties corrected the process for both centering and control. This changed the operation of the $\bar{X}$ chart centering and control limits, requiring the Q.E. to update the control chart based on recent data.

Thirty new subgroups of $n = 4$, taken after correction of the process, resulted in $\bar{R} = 208.70$, $UCL_R = 476.25$, $LCL_R = 0.00$, $\bar{\bar{X}} = 1,228.45$, $UCL_{\bar{X}} = 1,380.59$, and $LCL_{\bar{X}} = 1,076.31$. The highest R value was 432 and the highest and lowest $\bar{X}$ values, respectively, were 1,354.75 and 1,099.50. No “X’s” were triggered by any of the four sensitivity checks. The process is now in a SOSC.

8.14. Process Capability Analysis

It is important for us to note that a process in control does not necessarily mean that all of the product will be good—or that *any* of it will be good. When a process is

⁶Adapted from *AT&T Statistical Quality Control Handbook*, 2nd ed. (Easton, Pa.: Mack Printing Co., 1984).

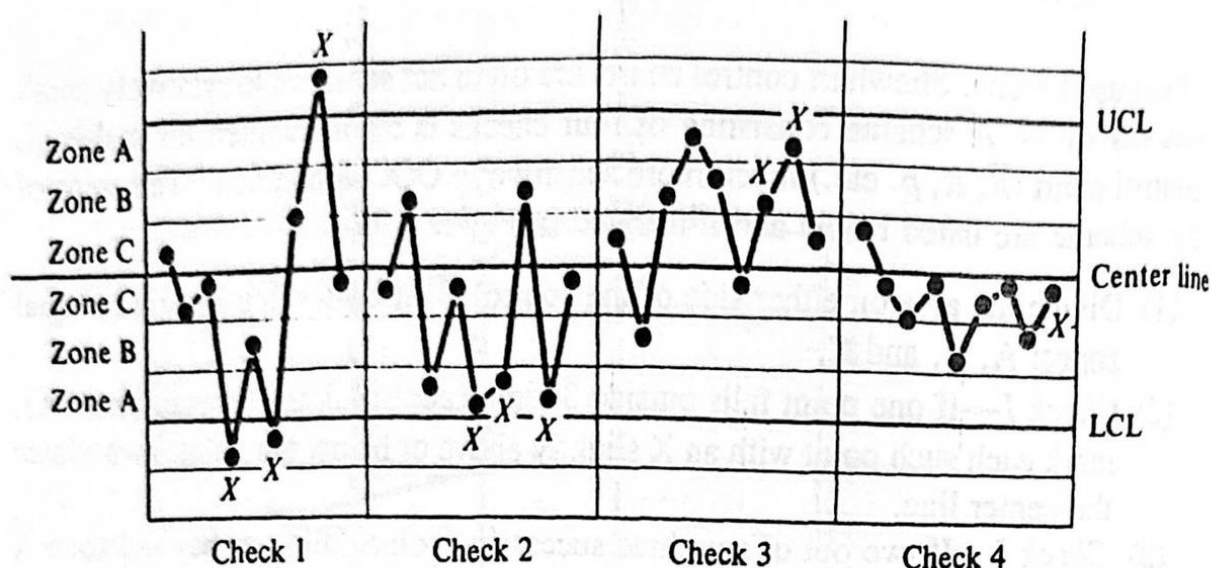


Fig. 8.17. Checks for sensitizing control charts.

in control, it simply means that a common-cause system is operating. It is possible, for example, to have a process in control far away from the desired specification limits. We must, therefore, assure ourselves that the process is not only in control, but that it is also *in control at the correct process centering or mean dimension*.

It is also possible for us to have an in-control process with perfect centering and still have an excessive fraction nonconforming. This can occur when we have both upper and lower specifications (U and L) that are too close when compared with the inherent variability of the process. This means that the process is operating as well as possible, but it is just not capable of producing all of the items within specifications. In this situation, we must either change processes, widen the design specifications, or be prepared to sort good items from bad.

From our control chart data, we can easily determine if the specifications U and L will be too narrow for our process. We know that nearly all (99.73% in the case of a normal distribution) of the items produced by a distribution will be within three standard deviations of its mean; that is, the natural variability of the process is $\bar{X} \pm 3\sigma$. We estimate the process standard deviation for an in-control process as follows:

$$\sigma = \frac{\bar{R}}{d_2} \quad (8.8)$$

where d_2 is a factor that depends on the subgroup size and is given in Table 8.1. Now, if 6σ exceeds $U-L$ when the process is in control, we know that the inherent variability of the process is simply too great.

Example 8.4

Your Q.E. can now estimate the inherent variability of the strength of super-gadgets based on recent, in-control data:

$$\sigma = \frac{\bar{R}}{d_2} = \frac{208.71}{2.059} = 101.36$$

$$\bar{\bar{X}} \pm 3\sigma = 1,228.45 \pm 3(101.36) = 924.37, 1,532.53$$

Since only a lower specification is applicable, the Q.E. cannot compare 6σ to $U-L$, but he can see that some fraction of product will still be nonconforming. Assuming a normal distribution, calculating $Z = (1,000 - 1,228.45)/101.36 = -2.25$, and using the standard normal table in Appendix B, it can be seen that the fraction nonconforming below a strength of 1,000 is now about 1.22%.

8.15. Control Charts for Attributes

8.15.1 The p Control Chart

The p chart is an attributes data chart based on the fraction of product nonconforming to specs. A subgroup of n items is selected from a process and each is inspected for nonconformities on multiple characteristics. Any item found to have one or more nonconformities is considered a nonconforming item. The number of nonconforming items x divided by the sample size n is the fraction nonconforming.

The p chart, like $\bar{X}$ and R charts, is very popular in industry. It requires only attributes data, so go-no go gauging, visual inspections, and the like yield appropriate data. The p chart is particularly attractive where combining several characteristics into one chart is desired, unlike $\bar{X}$ and R charts. Unfortunately, common subgroup sizes are often in the range of 50 to 300 items.

Approximately $m = 20$ or 30 subgroups of size n are selected from the process, keeping in mind the desire for within-subgroup homogeneity, with production process differences, if any, showing up over time between subgroups. For each subgroup i , the statistic $p_i = x_i/n_i$ is calculated.

We recall that the number of nonconforming items in a sample of size n from an infinite process is described by the binomial distribution with mean $\mu = np$ and variance $\sigma^2 = npq$, where $q = 1 - p$ and where p is the process fraction defective. Since we are interested in a p chart, the appropriate measure for each subgroup is x/n , or the subgroup fraction nonconforming which has mean $\mu = p$ and variance $\sigma^2 = pq/n$. Thus, 3σ control limits are established from m subgroups by using the following equations:

$$\text{center line} = \bar{p} = \frac{\sum_{i=1}^m x_i}{\sum_{i=1}^m n_i} \quad (8.9)$$

$$\text{upper control limit} = \text{UCL}_p = \bar{p} + 3 \sqrt{\frac{\bar{p} \bar{q}}{n_i}} \quad (8.10)$$

$$\text{lower control limit} = \text{LCL}_p = \bar{p} - 3 \sqrt{\frac{\bar{p} \bar{q}}{n_i}} \quad (8.11)$$

The center line and control limits are plotted on graph paper, and the individual statistics p_i are plotted in chronological order. The same four-check sensitivity scheme presented earlier may be applied.

Two additional items are important to note. First, the subgroup size n_i may not be constant from subgroup to subgroup on the p chart. In this case, individual control limits for each point may be calculated using Equations (8.10) and (8.11) with the appropriate subgroup sizes inserted for n_i . Second, it may be desired for later troubleshooting to keep a record of which characteristic nonconformities are observed during inspection, even though a knowledge of specific nonconformities is not needed for the p chart.

Example 8.5

Your Q.E. is now in the habit of achieving successes using SPC tools to diagnose processes. A decision is made to utilize a p chart in-house on the good-gadget line. The data in Table 8.3 are collected over a 3-week period.

The center line and control limits are calculated using Equations (8.9), (8.10), and (8.11) as follows:

$$\begin{aligned}\bar{p} &= \frac{263}{2,885} = 0.0912 \\ \text{UCL}_p &= 0.0912 + 3 \sqrt{\frac{(0.0912)(0.9088)}{n_i}} \\ &= 0.1775 \text{ at } n = 100 \\ &= 0.2026 \text{ at } n = 60 \\ &= 0.2639 \text{ at } n = 25 \\ \text{LCL}_p &= 0.0912 - 3 \sqrt{\frac{(0.0912)(0.9088)}{n_i}} \\ &= 0.0048 \text{ at } n = 100 \\ &= 0.0 \text{ at } n = 60 \text{ and } n = 25\end{aligned}$$

A p chart for the data is shown in Figure 8.18. The three X's indicate that the good-gadgets line was not operating in a SOSOC. The Q.E. finds that subgroup 7 was a recording error and subgroups 13 and 19 were due to an untrained operator. Also, a separate list of nonconformities clearly showed that most nonconforming items were due to scratches caused by improper material handling. The Q.E. quickly instituted a training program on material handling for all operators, and a comprehensive training program for occasional substitute operators.

8.15.2 The c Control Chart

The c chart is based on the number of nonconformities observed per subgroup. The size of each subgroup must first be established and can, for example, consist of an airplane skin section, five stereo receivers, three forklift trucks, 10 pairs of overalls, and so on. The size of the subgroup must remain fixed from subgroup to subgroup in order to maintain constant the area of opportunity for the occurrence of nonconformities. The c chart, like the p chart, requires only attributes data. It

Table 8.3. SAMPLE DATA FROM GOOD-GADGET LINE

Subgroup Number	Subgroup Size	Number Nonconforming	Fraction Nonconforming
1	100	9	0.09
2	100	4	0.04
3	100	6	0.06
4	100	11	0.11
5	100	13	0.13
6	60	12	0.20
7	100	0	0.00
8	100	8	0.08
9	100	15	0.15
10	100	12	0.12
11	100	8	0.08
12	100	6	0.06
13	100	18	0.18
14	100	9	0.09
15	100	7	0.07
16	100	11	0.11
17	100	8	0.08
18	100	8	0.08
19	25	9	0.36
20	100	10	0.10
21	100	7	0.07
22	100	3	0.03
23	100	12	0.12
24	100	4	0.04
25	100	6	0.06
26	100	7	0.07
27	100	10	0.10
28	100	4	0.04
29	100	11	0.11
30	100	15	0.15
	2,885	263	

is attractive where a simple *count* of nonconformities (deviant visuals, speeds, frequencies, dimensions, etc.) is the performance measure of interest.

Approximately $m = 20$ or 30 subgroups are selected over time. The items selected for a subgroup should be homogeneous in the sense that they are produced together or under similar conditions. Once again, production process differences, if any, should show up over time between subgroups. For each subgroup, the statistic c_i is kept, representing the total count of nonconformities observed over all units in the subgroup. The c chart center line and control limits are given as follows:

$$\text{center line} = \bar{c} = \frac{\sum_{i=1}^m c_i}{m} \quad (8.12)$$

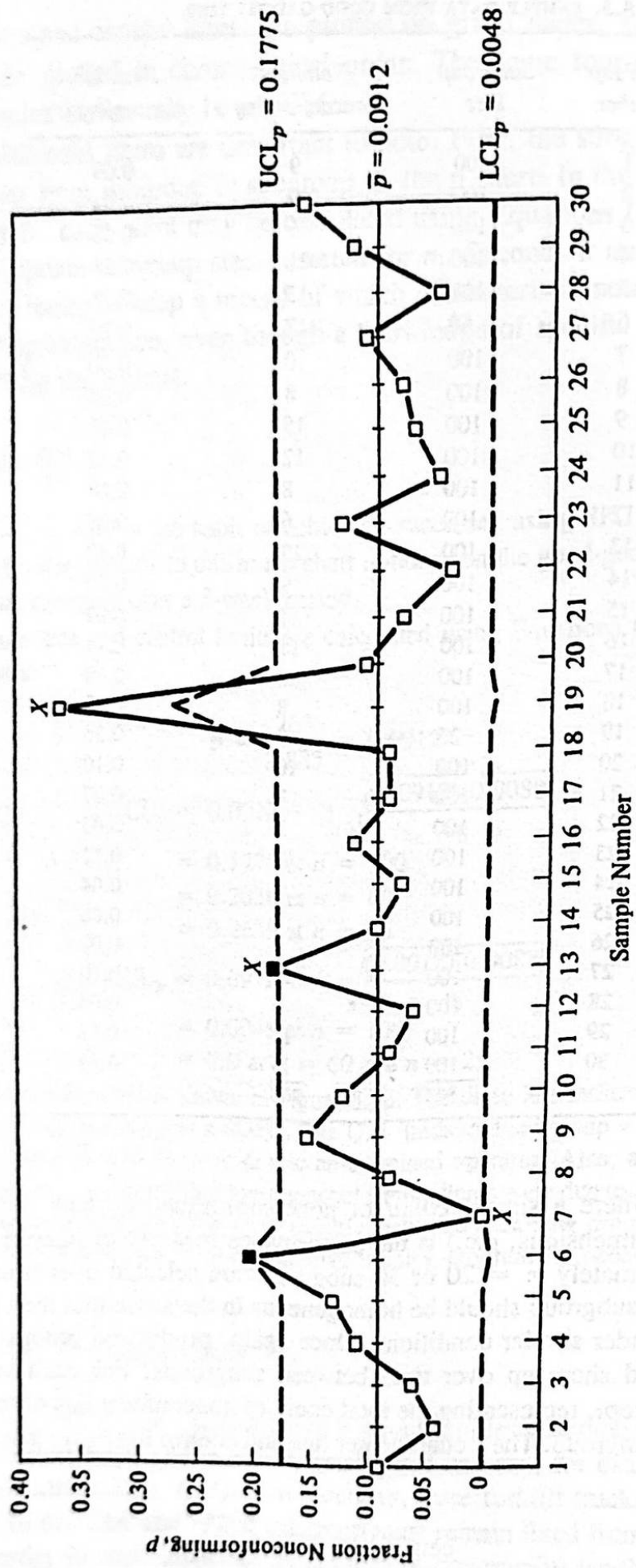


Fig. 8.18. p chart for good-gadget line.

$$\text{upper control limit} = \text{UCL}_c = \bar{c} + 3 \sqrt{\bar{c}} \quad (8.13)$$

$$\text{lower control limit} = \text{LCL}_c = \bar{c} - 3 \sqrt{\bar{c}} \quad (8.14)$$

The center line and control limits are plotted on graph paper, and the individual statistics c_i are plotted in chronological order. The same four-check scheme discussed previously is applicable.

Example 8.6

You now have a new dilemma. Your Q.E. has learned that an industrial engineering degree coupled with a knowledge of SPC is in high demand. The vendor has just made the Q.E. an offer to come to work as quality manager at a 30% increase in salary. What are you going to do as president of Gadgets, Inc.?